B.Sc Honours – Physics

Empowering Students with Scientific Knowledge & Research Excellence

The **B.Sc Honours – Physics** programme at **St. Ann's College for Women** is a four-year undergraduate programme designed to provide strong foundations in classical and modern physics, analytical thinking, experimental techniques, computational skills, and scientific research.

The programme helps students understand the principles governing matter, energy, motion, electricity, optics, electronics, quantum physics, nanotechnology, and renewable energy systems. Along with theoretical knowledge, students gain extensive practical exposure through laboratory experiments, computational physics, and project-based learning.

Physics develops logical reasoning, problem-solving ability, scientific temperament, and innovation skills, preparing students for careers in research, education, technology, electronics, data science, renewable energy, and higher studies.

Note: St. Ann's College for Women presently offers the B.Sc Honours Physics programme as a Three-Year Degree Programme in accordance with the Acharya Nagarjuna university and government guidelines.

Programme Highlights

- Strong foundation in Classical & Modern Physics
- Advanced Physics Laboratory Training
- Practical Exposure in Electronics & Instrumentation
- Computational Physics using Python & Numerical Methods
- Renewable Energy & Nano Science Applications
- Research-oriented Curriculum
- Skill Development through Practical Experiments
- Industry & Technology-based Learning
- Opportunities for Higher Studies & Research
- Career Guidance & Competitive Exam Preparation

Core Subjects Offered

Semester-wise Major Subjects

- Introduction to Mathematical Physics
- Mechanics and Properties of Matter
- Waves and Optics
- Heat and Thermodynamics
- Atomic, Molecular & Nuclear Physics
- Basic Electronics
- Applied Optics
- Electricity, Magnetism and Electromagnetic Theory
- Analog Electronics
- Advances in Physics
- Introduction to Solid State Physics
- Electronic Instrumentation
- Solar Energy and Applications
- Fundamentals of Nano Science
- Solar, Thermal and Photovoltaic Conversion
- Fundamentals of Python and Numerical Methods
- Electronic Devices and Circuits
- Low Temperature Physics and Refrigeration
- Synthesis of Nano Materials
- Wind, Hydro, Ocean & Geo-thermal Energy
- Computations in Mechanics, Waves and Oscillations
- Analog and Digital Electronics
- Vacuum Technology
- Characterization of Nano Materials
- Energy Storage and Conversion Systems

- Computations in Optics, Heat and Thermodynamics
- Electronic Communication Systems
- Materials for Industrial Applications
- Applications of Nanomaterials
- Biomass and Hydrogen Energy
- Computations in Electricity, Magnetism, Electromagnetic Theory and Modern Physics

Practical Laboratory Courses

The programme includes extensive practical training in:

- Physics Laboratory Experiments
- Optics & Wave Experiments
- Electronics & Circuit Design
- Electrical Measurements
- Nano Science Experiments
- Renewable Energy Applications
- Computational Physics using Python
- Instrumentation Techniques
- Communication Systems
- Modern Physics Experiments

Minor Course – Computer Science

To enhance digital and technical skills, students can choose **Computer Science** as a Minor Programme.

Minor Subjects Include

- Computer Fundamentals and Office Automation
- Problem Solving Using C
- Database Management System
- OOPS Through JAVA
- Python Programming
- Web Interface Design Technologies

Practical laboratory training is provided for all Computer Science minor subjects.

Career Opportunities

Graduates of B.Sc Honours – Physics can pursue careers as:

- Physicist
- Research Assistant
- Scientist
- Lab Technician
- Electronics Technician
- Radiation Assistant
- Data Analyst
- Statistician
- Physics Teacher
- Technical Assistant
- Research Associate
- Instrumentation Technician

Higher Education Opportunities

Students can pursue:

- M.Sc Physics
- M.Sc Applied Physics
- M.Sc Electronics
- M.Sc Nuclear Physics
- M.Sc Atmospheric Sciences
- M.Tech Programmes
- Integrated M.Sc Programmes
- Ph.D in Physics & Applied Physics
- MBA & Interdisciplinary Programmes

Top Recruiters

Students may find opportunities in:

- Indian Space Research Organisation (ISRO)
- Defence Research and Development Organisation (DRDO)
- Bhabha Atomic Research Centre (BARC)
- National Institute of Oceanography (NIO)
- Research Laboratories & Universities
- Electronics & Renewable Energy Industries
- Educational Institutions

Eligibility

Students who have completed Intermediate / 10+2 with **MPC or BiPC** from a recognized board are eligible for admission.

Why Choose Physics with Computer Science Minor at St. Ann's?

The combination of **Physics Major with Computer Science Minor** provides students with both scientific knowledge and modern technological skills, creating excellent career opportunities in research, software, electronics, data science, artificial intelligence, and emerging technologies.

At **St. Ann's College for Women**, students gain interdisciplinary learning through Physics, Programming, Computational Methods, Electronics, Python, Database Management, and Web Technologies.

Benefits of this Combination

- Strong foundation in Physics and Computational Science
- Programming Skills using C, Java & Python
- Exposure to Data Analysis & Scientific Computing
- Knowledge in Electronics & Modern Technology
- Better Career Opportunities in IT & Research Fields
- Preparation for Artificial Intelligence & Data Science Careers
- Skill Development in Problem Solving & Logical Thinking
- Excellent Foundation for Higher Studies & Competitive Exams
- Industry-oriented and Future-ready Curriculum

This combination helps students become highly employable in both scientific and technological sectors.

Contact Us

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